Cluster Analysis of COVID-19

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➢ Executive Summary

During the COVID-19 outbreak between early 2020 and mid-2023, Texas lawmakers had to make difficult decisions to mitigate the virus's impact while balancing public health and economic concerns. This project analyzes the effectiveness of two primary interventions—social distancing measures and vaccination campaigns—in reducing infection rates and mortality. By clustering county-level data, we assess how different regions were affected and identify similarities to help customize a response for their specific needs. The findings highlight that while social distancing initially helped slow infection rates, its effects were temporary and often reversed once restrictions were lifted. In contrast, vaccination had a more sustained impact, significantly lowering hospitalizations and deaths. The results suggest that future pandemic responses should prioritize rapid vaccine distribution, booster campaigns, and healthcare accessibility over prolonged social distancing measures to clustered counties. By using real-time data to study the changes of COVID-19, Texas can ensure a more effective and sustainable approach to managing future health crises.

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1 Problem Description

COVID-19, caused by the SARS-CoV-2 virus, is highly contagious and has caused an international public health crisis. The virus is transmitted primarily through respiratory droplets and its symptoms can be mild flu-like manifestations to severe respiratory distress and death. Due to its rapid rate of spread and the fact that its variants are constantly evolving, COVID-19 has placed immense pressure on healthcare systems all over the world. To address how we can limit the spread and keep the population healthy, Texas policymakers should consider the following questions:

- How has social distancing affected COVID-19 infection rates compared to the impact of vaccination campaigns? Having knowledge of the impact of each intervention on infection rates will tell policymakers whether in the future policy should concentrate on distancing interventions, vaccination campaigns, or both.
- Which of the two interventions—social distancing or vaccination—had greater impact in reducing mortality? This is crucial to optimize resource utilization and avoid overloading healthcare systems.

These policymakers must balance protecting public health with economic and social stability. With limited healthcare resources, such as hospital beds and medical personnel, state officials must use data to allocate assistance effectively, with high-risk areas receiving the support they require. This project aims to provide key data that will benefit state-level decision-makers to improve emergency response plans as well as inform well-informed, evidence-based public health policies.

2 Data Collection and Quality

➢ Data Gathering

All data used in this report is sourced from the same sources in the last report.

- Google Cloud Public Marketplace, which provides many publicly available datasets hosted on Google Cloud. These datasets cover a wide range of topics, including data on COVID-19, economic indicators and geographic information.
- Google COVID-19 Open Data Repository, which is a dataset compiled by Google, aggregating COVID-19 data from multiple global sources. The data for Texas is supplied by the Texas Department of State Health Services.
- CDC (Centers for Disease Control and Prevention) COVID-19, which has official data reported by public health agencies.

➢Data Quality and Preparation

The data is once again combined at the county level using the unique FIPS code, for clustering, we will not be using generalized data from the state level since it will not provide much insight outside of state-wide trends that have already been explored previously.

The data has already been cleaned, and names have been standardized for smooth joining. All the dates will follow a yyyy-mm-dd format, and data that has been aggregated into weeks will not be used in clustering analysis. This is because weekly aggregation was used for a continuous timeline showing the general trend in state level virus control. There are not any outliers since they can all be attributed to real-life events, such as new variants or interventions.

For clustering, the data needs to be processed even further. First, variables that are not numerical must be removed from the table that will be clustered since scaling requires numbers only. Then not applicable entries also must be removed, as these will not scale. By scaling the variables, each cluster will have a Z-score showing how much that variable in the cluster will deviate from the norm, which is zero.

To have higher accuracy in the clustering, I needed to remove some other variables that were previously here. Retail_and_rec_pct was problematic because it had many NA values throughout the data set and was giving very poor clusters as a result. Many counties were removed, and that created problems, so it needed to be resolved. I decided to either impute the missing values or exclude this feature entirely, allowing for more consistent data and better-defined clusters that could provide more meaningful insights. This adjustment helped improve the clustering performance and the overall interpretation of the visual results.

Feature	Scale of Measurement	Description
Total_pop	Ratio	The total population in that specific county
Median_income	Ratio	Median income of individuals living in the county
Median_age	Ratio	Median age of individuals living in the county
cases_per_1000	Ratio	Out of every 1000 people, how many were infected by COVID-19
deaths_per_1000	Ratio	Out of every 1000 people, how many died from COVID-19
death_per_case	Ratio	The ratio of people dying to those infected
First_Dose_Pct	Ratio	People with at least one Dose by State of Residence
Full_Dose_Pct	Ratio	Percent of people who have completed a primary series (have second dose of a two-dose vaccine or one dose of a single-dose vaccine) based on the jurisdiction and county where vaccine recipient lives
Full_Dose_Pct65Plus	Ratio	Percent of people ages 65+ who have completed a primary series (have second dose of a two-dose vaccine or one dose of a single-dose vaccine) based on the jurisdiction where vaccine recipient lives
Work_pct	Interval	Mobility trends for showing the percentage change of people coming into work

Figure 1: Feature Descriptions.

3 Data Exploration

To effectively streamline the process, four timestamps will be taken to observe the trends without being overwhelmed. These timestamps are hand picked to large events in the timeline of COVID-19, showing us causes and effects of decisions made by the United States policymakers.

- **2020-07-15:** The first major infectious wave, also known as the "Summer Wave." WHO officially recognized the possibility of airborne transmission just one week earlier. Many countries and U.S. states began implementing mandatory mask policies as the CDC and WHO increased their guidance on wearing masks in public spaces
- **2021-10-22:** Marks a key date in the U.S. vaccine rollout, as the FDA authorized booster doses for Moderna and Johnson & Johnson vaccines. Although vaccines have been released to the public for almost a year, we can see pre-booster trends, to identify the power of the initial vaccines. This also makes it more difficult to pinpoint the effectiveness of keeping individuals healthy and safe as there is now more than one variable at play. By taking this timestamp, we can evaluate the before and after trends to see how effective it is.

These time stamps highlight a dynamic interaction between the virus' spread, public health interventions, and vaccine efforts. By focusing on these key moments, we can obtain a comprehensive understanding of how strategies like mask mandates, vaccinations, and boosters shaped the course and outcome of the pandemic.

➢ Statistical Summary

Table 2 gives the statistical overview of key numeric variables in the data set, providing information regarding population characteristics, vaccination rates, mobility trends, and COVID-19 impact. Categorical variables like county name and state names were not accounted for in the overview. Total population varies widely across counties from 1,084 to 4,525,519 with a very high standard deviation of 399,834.4. This demonstrates the vast variation between urban and rural populations. The mode of 4,525,519 is likely to be the largest county duplicated because there are no exact copies elsewhere and therefore skews the distribution.

Median income also shows considerable variation across counties, from $68,851 and with a standard deviation of 11,619. This economic disparity is also a contributing factor to disparate public health outcomes among areas. By age, the median varies from 25.8 to 57.1 years, showing age diversity of county populations that can affect vaccine prioritization as well as health outcomes.

On the vaccine front, first dose and full dose percentages have wide ranges (77.6% and 76.7%, respectively), indicating mixed rates over time and across counties. The modal value of 95% on both measures indicates a good final surge to complete vaccination coverage. For 65+ year olds, full vaccination is similarly high, with a third quartile value of 81.6%, indicating prioritizing the vulnerable population. Mobility patterns, such as workplace movement, have high variability.

Finally, COVID-19 case and death rates per 1,000 residents highlight the disproportionate impact of the pandemic. The case rate varies between 23.11 and 182.90 with a standard deviation of 25.95, and the death rate varies between 0.30 and 5.46, showing enormous differences in exposure and outcomes. The death-to-case ratio varying between 0.0038 and 0.0932 also highlights the disproportionate severity of outcomes across counties.

Variable	Min.	1st Qu.	Median	3rd Qu.	Max.	Mode	Std Dev	Range
Total_pop	1084	8082	20082	53171	4525519	4525519	399834.4	4524435
Median_income	24794	42354	47923	54971	93645	44601	11619.06	68851
Median_age	25.80	34.60	38.35	42.90	57.10	35.6	5.839018	31.3
First_Dose_Pct	22.30	44.40	50.30	59.90	99.90	95	13.43828	77.6
Full_Dose_Pct	18.30	38.60	43.90	51.80	95.00	95	11.6431	76.7
Full_Dose_Pct65 Plus	15.70	67.20	73.80	81.60	99.90	95	10.79269	84.2
Retail_and_rec_ pct	-95.00	-19.00	-9.00	-1.00	197.00	-10	16.44832	282
Work_pct	-89.0	-31.0	-23.0	-15.0	40.0	-18	13.76627	129
cases_per_100 0	23.11	59.43	75.75	94.24	182.90	63.2758 4	25.95004	159.790 9
deaths_per_10 00	0.2987	1.2191	1.7347	2.3794	5.4608	0.84520 69	0.922084	5.16211 6
death_per_cas e	0.003789	0.0153 99	0.0228 40	0.0304 73	0.09322 0	0.02347 418	0.0121769 8	0.08943 136

Figure 2: Statistical Summary.

4 Modeling and Evaluation

➢ Determining Number of Clusters

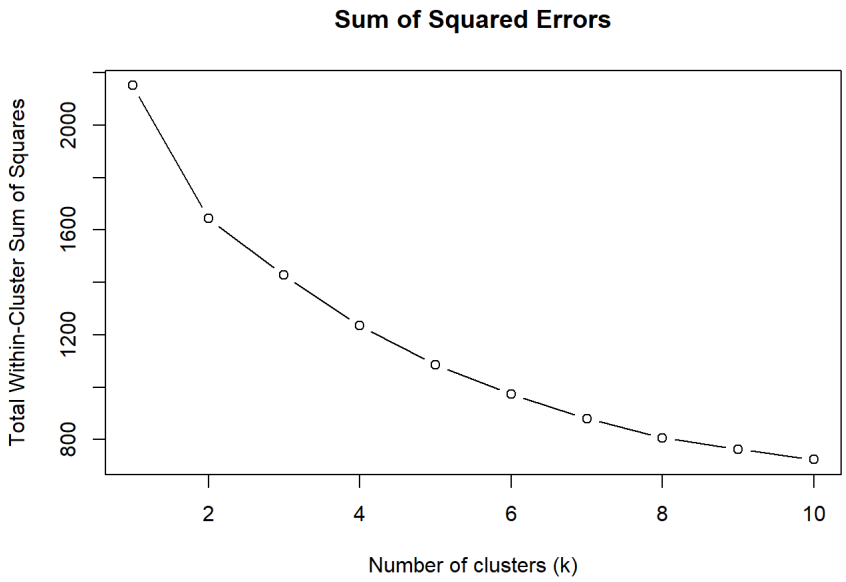

Figure 3: Sum of Squared Errors

To determine the best number of clusters, I use the Sum of Squared Errors to see the optimal value for k. The optimal value will be the point whereas the SSE decreases, the change in slope from point to point flattens out. Here, point 3 was determined to be the strongest number of clusters because after 3, the performance begins to flatten out and is the turning point of significant gain. So, for all further clusterings, there will always be three groups for k-means clustering, and the hierarchical clustering will be cut off at three clusters.

➢ K-Means Clustering

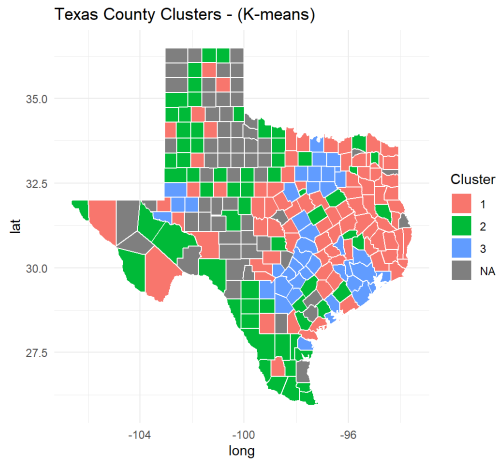

Figure 4: K-means on social distancing county map

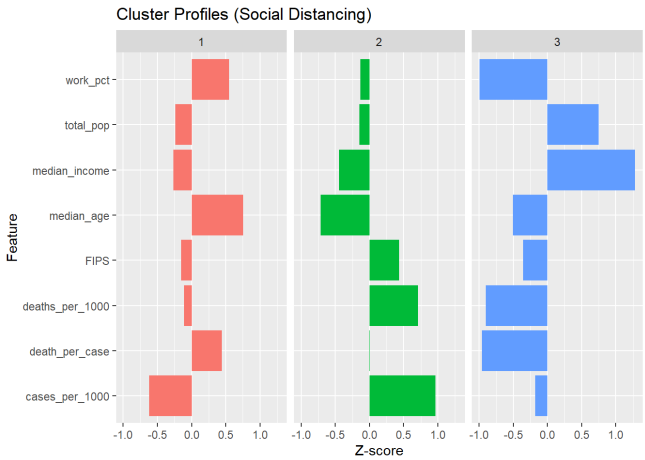

Figure 5: K-means on social distancing cluster profiles

The map illustrates the results of k-means clustering applied to Texas counties based solely on social distancing behavior prior to the availability of COVID-19 vaccines. This map shows only data taken before vaccines

and boosters were mass produced. Counties are grouped into three distinct clusters, each represented by a different color while grey counties indicate missing or unavailable data. This missing data is due to the lack of reporting mobility data early in the lockdown and k-means clustering requiring the handling of NA values. The spatial distribution shows regional patterns in social distancing adherence. Blue clusters are around the higher populated areas, while the green is more akin to the western counties and the red is more about the eastern counties. This clustering provides insights into how different parts of Texas responded behaviorally to early public health recommendations, reflecting regional differences in social distancing practices.

This bar plot presents the standardized (z-score) feature values for each of the three clusters identified through K-means. The Z-score shows how far off the average that cluster is in that category. By an analyzing the graph, we can note that the red cluster shows an increase in working hours away from home, has more older people than across the other counties, and has a much larger decrease in cases per 1000. The green plot values a lower median age, sees higher deaths per 1000, and has a much higher cases per 1000 than the other two clusters. The blue cluster has a major decrease in working, a much higher median income, and a significantly lower death and case count per 1000.

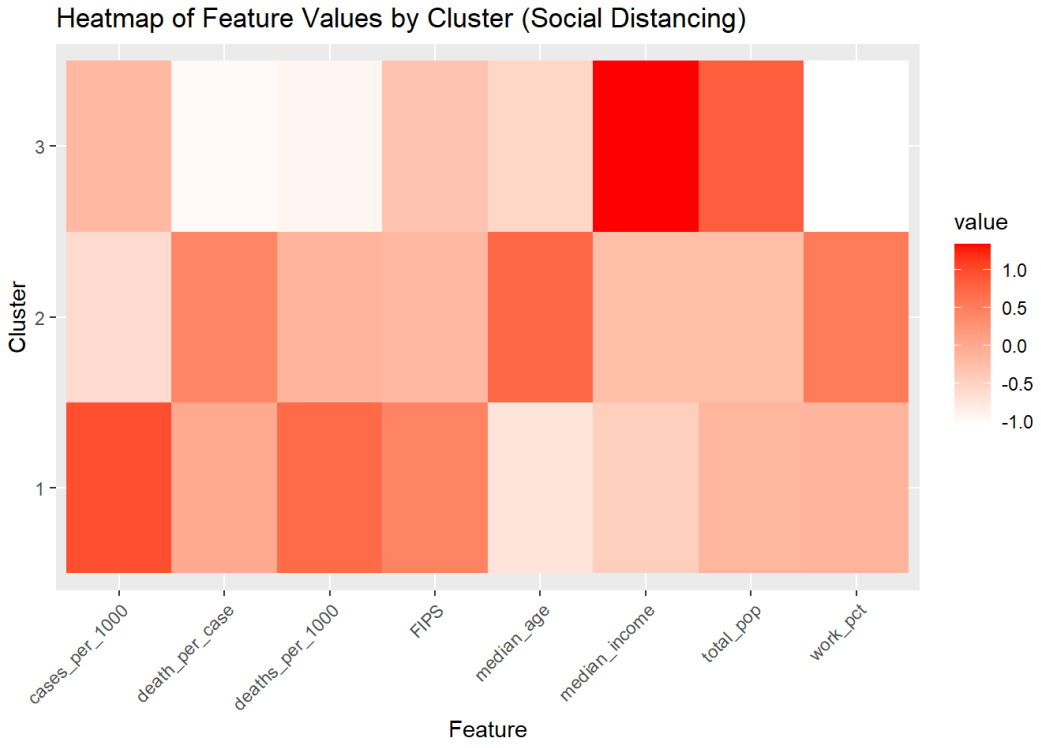

Figure 6: Heatmap on feature importance to each cluster pre-vaccines

This heatmap shows the z-score feature values of each of the three clusters for the Texas counties at the pre-vaccine time during the COVID-19 pandemic period. Red represents greater-than-average value, while light color represents less-than-average values for a feature. Cluster 1 is defined by consistently high values across health

indicators, including cases per 1,000, deaths per 1,000, and death per case, showing that this cluster of counties experienced the poorest pandemic. Cluster 2 has relatively moderate values in all features but is marginally higher on death per case and work_pct, perhaps showing places with essential workers and little mobility loss. Cluster 3, conversely, possesses relatively neutral or low values for every health measure but peaks wildly high on median_income and total_pop, which implies wealthier and more populous counties with better health outcomes, likely due to their having access to more resources or stricter social distancing adherence. Overall, the heatmap illustrates that the COVID-19 socioeconomic and health metrics varied within the clusters. This assists with the interpretation of behavioral and structural heterogeneity within Texas counties at the beginning of COVID-19 mitigation.

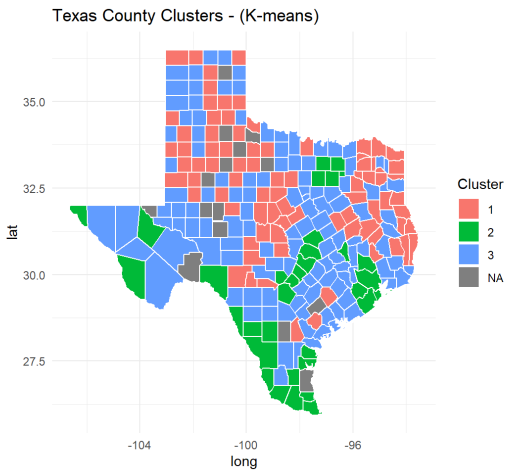

Figure 7: K-means on social distancing county map

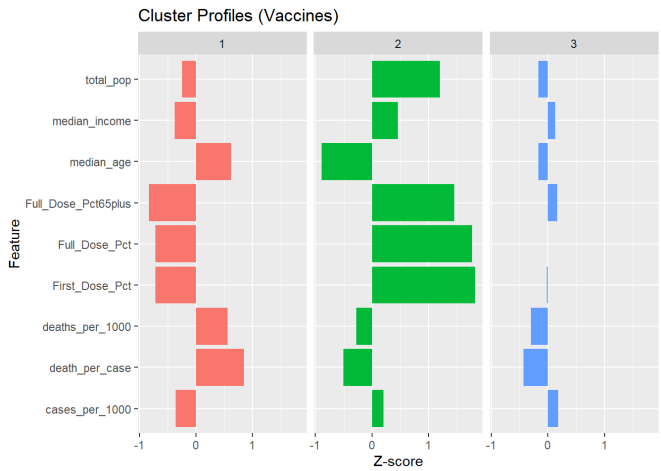

Figure 8: K-means on social distancing cluster profiles

The map illustrates the results of k-means clustering applied to Texas counties based on how the public responded to the introduction of the vaccine and its boosters. Although we should take social distancing into consideration, it was left out of the diagram to focus in on the effect of vaccines. The spatial distribution shows regional patterns in social distancing adherence. Green clusters are around the higher populated areas, while red and blue clusters are more all over the place. This clustering provides insights into how different parts of Texas responded behaviorally to early public health recommendations, reflecting regional differences between the urban and rural areas of Texas.

By an analyzing the z-score bar graph, we can see that the red cluster has much less vaccinations compared to the other two, this can be because of lack of resources and distribution of treatment, or the residence of these counties purposefully avoiding the vaccine. Whatever the case may be, they show the only higher than normal deaths per case. The green cluster represents the urban areas, or counties that share some qualities with them. Most notably for the green cluster is the vaccination rates. People here are much more willing or have much easier access to the vaccine. There is a correlation with the green and blue having positive or zero z-score on vaccinations and having lower than average cases and deaths across the board. The blue cluster in its entirety depicts the average

county, it has no extreme z-scores in either direction, or I believe this is the case because of the drastic z-scores of the other two clusters, making this one a more intermediate approximation.

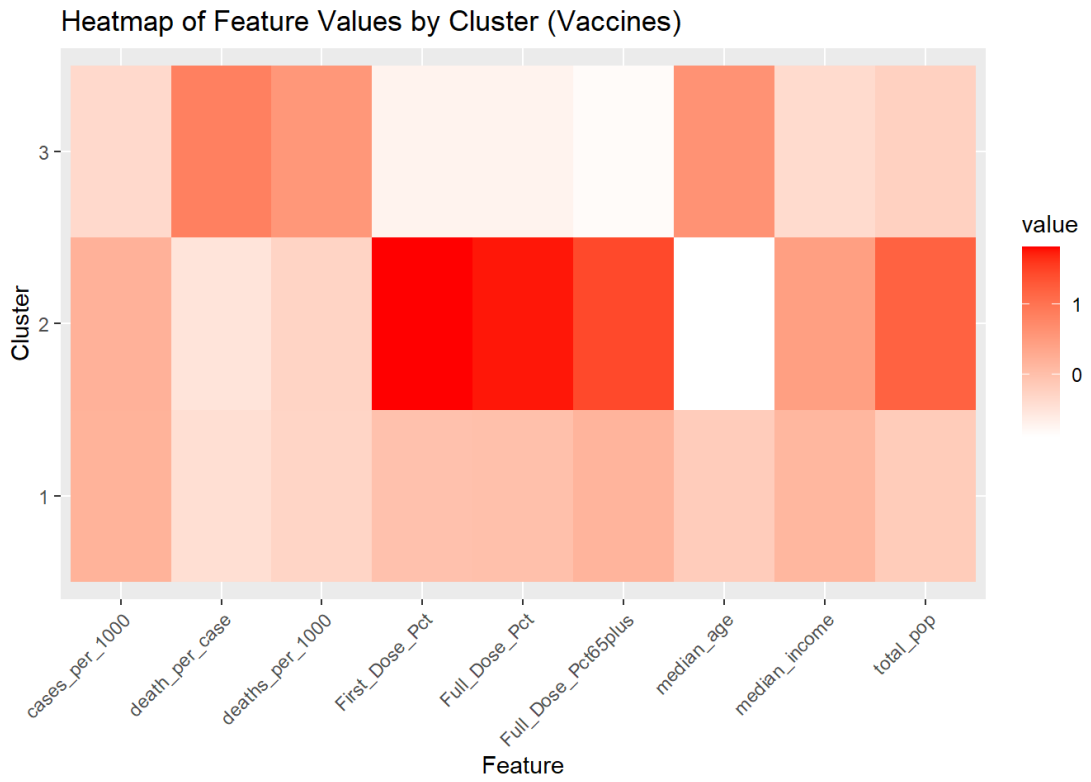

Figure 9: Heatmap on feature importance to each cluster post-vaccines

This heatmap is slightly less interesting than the pre-vaccination heatmap, since clusters 1 and 3 have no extremities that the clusters examined besides cluster 3 having a higher death rate compared to the other two. The most notable graphic on this heatmap is the extremely high value for receiving doses in the second cluster, meaning that the major feature that k-means clustered by was the vaccination rates across the counties, while a supplementary clustering was based on the death rate, as seen in cluster 3.

➢ Hierarchical Clustering

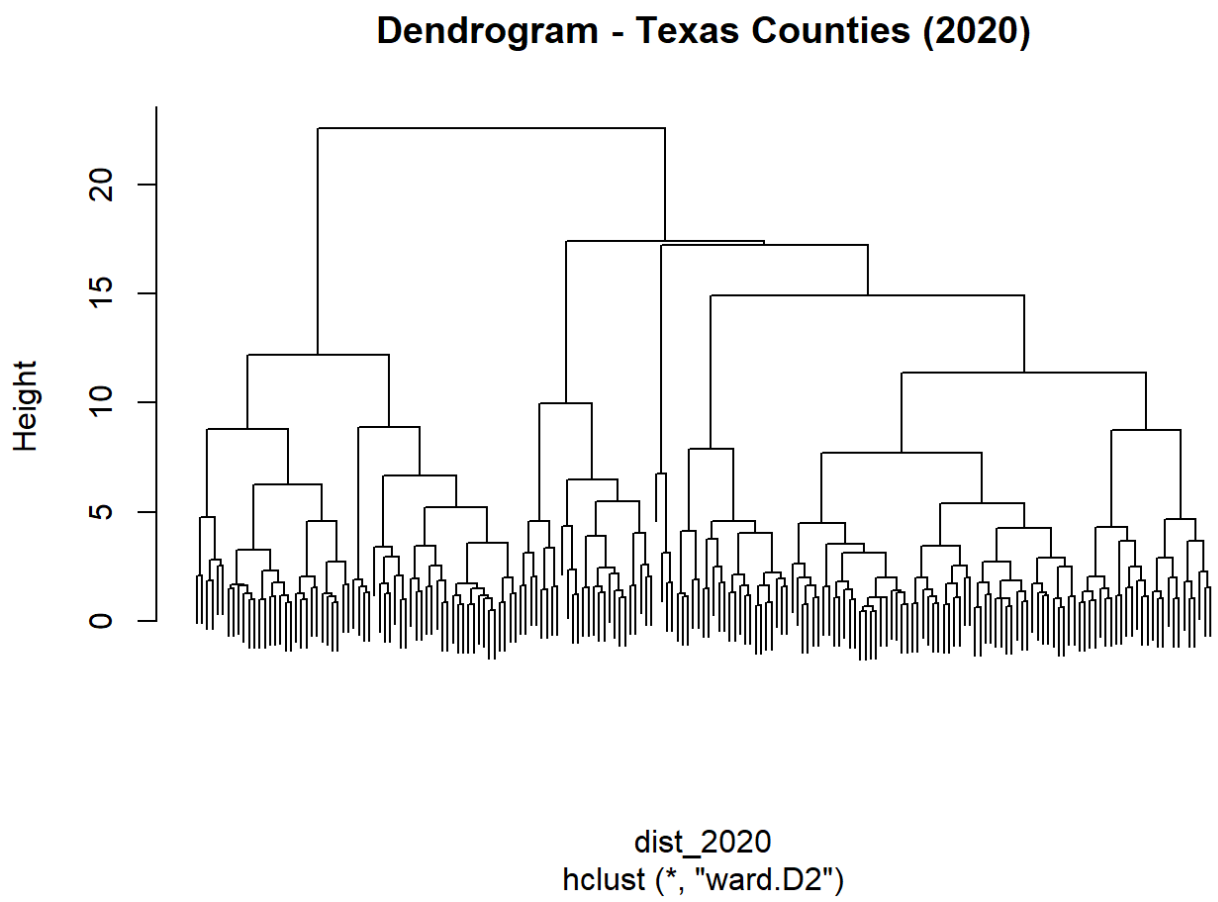

Figure 10: Hierarchical Clustering Dendrogram pre-vaccinations

The dendrogram illustrates the hierarchical clustering of Texas counties based on COVID-19 related variables during 2020, before widespread vaccination. This tree-like visualization shows how counties are grouped together based on similarity across multiple dimensions including demographic factors, infection rates, death rates, and mobility patterns. The vertical axis (Height) represents the dissimilarity distance between clusters - the higher the connection between branches, the more dissimilar the connected clusters are. The counties are arranged along the horizontal axis, with each vertical line at the bottom representing an individual county. The clustering was performed using Ward's D2 method (as indicated at the bottom of the figure), which minimizes the variance within clusters. This method tends to create more balanced, similarly-sized clusters compared to other hierarchical clustering approaches.

All the clusters seem normal besides the one in the very middle, which directly branches onto the others with a tiers connection. These could possibly be the urban counties being extremely different than all of the other counties in Texas, making them very dissimilar.

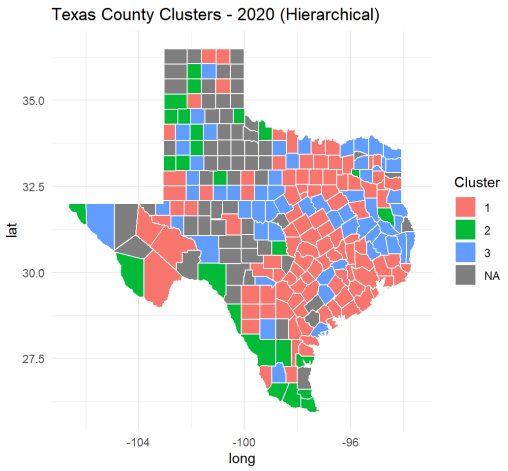

Figure 11: Hierarchical on social distancing county map

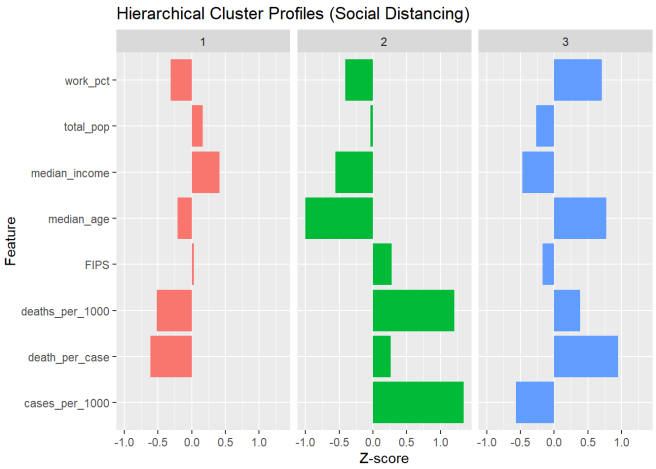

Figure 12: Hierarchical on social distancing cluster profiles

The hierarchical clustering took a different approach than the K-means clustering, where we saw the rural counties more intertwined in their clustering, this model seems to cluster the urban and its surrounding areas as the normal, and the rural counties show more drastic variance. The red cluster, which includes the urban cities, shows the lower deaths per case, while the green cluster seems to have incredibly high and extreme values for their deaths per 1000 and cases per 1000, while only being just a few counties. I believe that the hierarchical clustering has chosen these to be the outliers, where the virus hit the hardest in Texas. The blue cluster shows an increase in work, and an above average age, which leads into having the highest death per case of the three clusters.

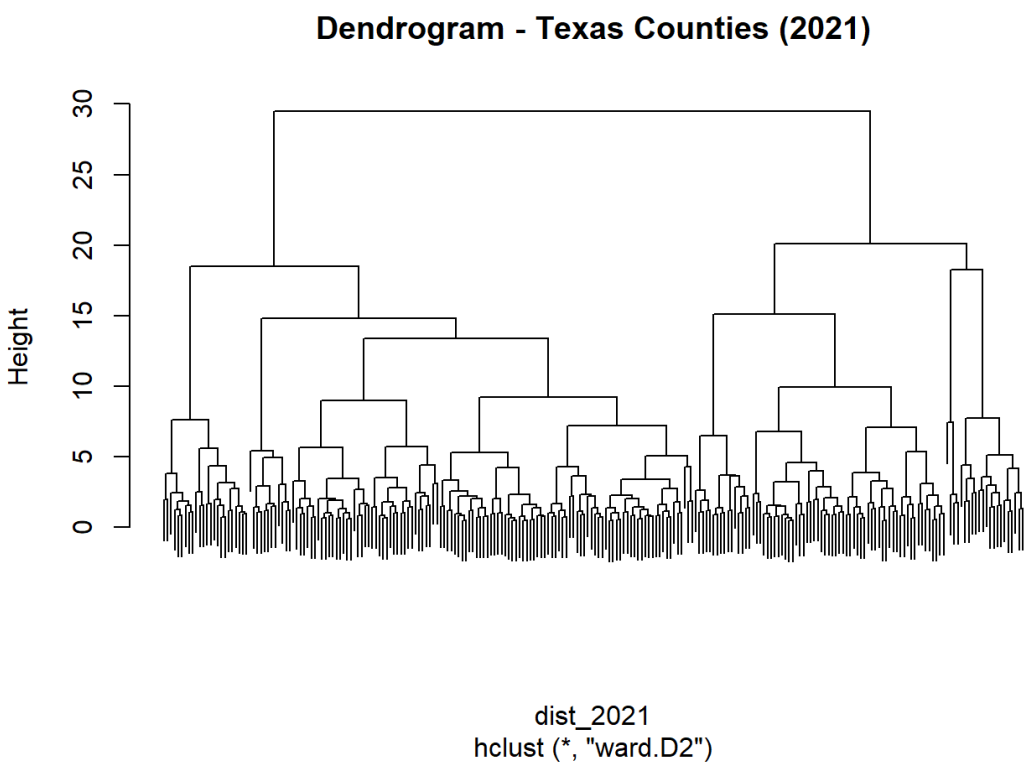

Figure 13: Hierarchical Clustering Dendrogram post-vaccinations

This dendrogram is much different compared to the first one, as it is much more uniform in its hierarchy structure. It clustered the urban counties together and some of the bordering counties to Mexico. By looking at the next figures (Figures 14 and 15), we can see that they are prioritized by vaccination rates more than anything else.

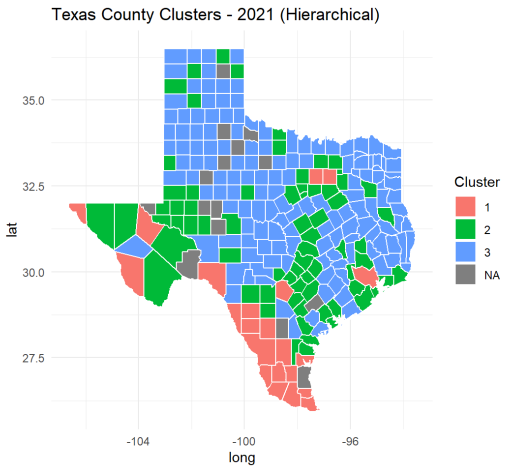

Figure 14: Hierarchical on social distancing county map (post-vaccine)

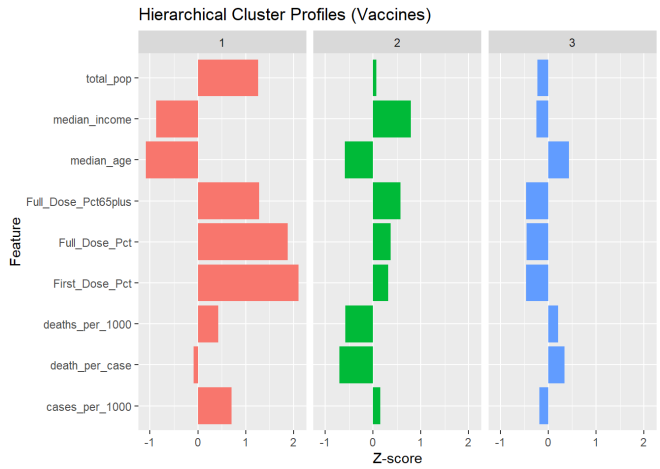

Figure 15: Hierarchical on social distancing cluster profiles (post-vaccine)

Here, the red counties represent high vaccination rates, not so much that they are urban. We can see that when clustering, the hierarchical model took the strategy of comparing vaccination rates, almost disregarding other metrics like deaths and cases. We normally have seen the urban clusters having a higher median income, but since it is clustering based on vaccination, the median income is much lower than normal, while blue and green counties have higher median incomes.

This shift in clustering highlights the dominant influence of vaccination rates on the hierarchical model, overshadowing other factors such as income levels and case mortality rates. As a result, the red counties, which are clustered together based on high vaccination rates, may not align with the typical urban-rural divide that we have observed in other analyses. The model appears to prioritize vaccination coverage as the primary characteristic, leading to a misalignment between the geographic or demographic traits we typically associate with high vaccination rates and other important health indicators. In contrast, the blue and green counties, which cluster around different vaccination metrics, retain higher median incomes, suggesting they represent the average with the red cluster including the extremes. This discrepancy underscores the need to reconsider how vaccination data is weighted in the clustering process, as it may not fully capture the complexities of public health outcomes across different communities.

➢Unsupervised Evaluation

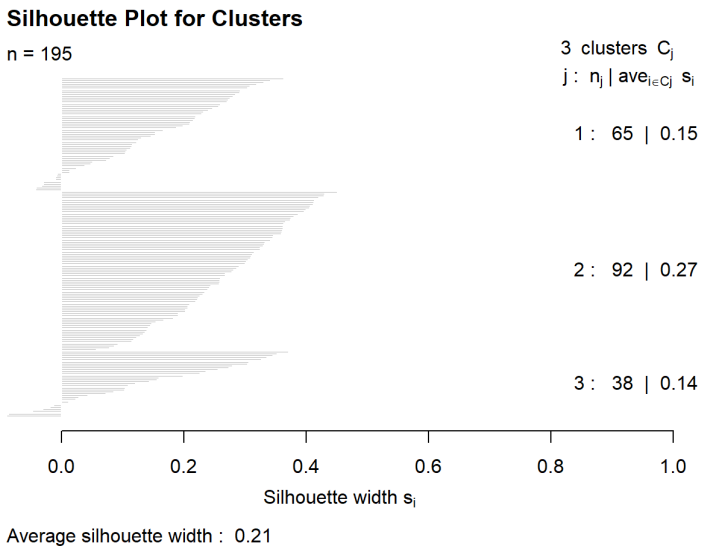

Figure 16: Silhouette Plot for Unsupervised Evaluation

From the silhouette plot analysis, we can infer the following about the clustering results: Cluster 1 consists of counties that are relatively similar to each other, but it also includes some points with negative silhouette scores. This suggests that while most of the counties in this cluster are well-grouped and exhibit good cohesion, there are a few counties that are not as well-matched to the cluster, potentially indicating either outliers or points that might have been assigned to the wrong cluster. Cluster 2, on the other hand, shows only positive silhouette scores, which indicates that all counties within this cluster are well-matched to each other and form a cohesive group with little to no misclassification. This cluster appears to be the most well-defined and depicts a strong relationship between its counties and separated from others, suggesting that the algorithm successfully captured a distinct set of counties. Cluster 3 shows mostly positive silhouette scores, but it has slightly more negative values than Cluster 1, indicating that while the majority of the counties are grouped well together, there are more counties that do not fit as well within the cluster. This could imply that Cluster 3 has a higher level of internal variability compared to the other clusters, or that some counties in this cluster might have been misclassified. Overall, the differences in silhouette scores between the clusters suggest that Cluster 2 is the most cohesive and Clusters 1 and 3 may need some work to get them more to be more unique from the other counties.

➢Supervised Evaluation

Figure 17: Cluster Distribution for Supervised Evaluation on Social Distancing

From this evaluation, we can assess the effectiveness of the clustering models in accurately capturing the varying levels of risk across counties and grouping them accordingly. The model's ability to differentiate between high- and low-risk counties is crucial in understanding how well it reflects the underlying factors contributing to mortality and health outcomes. In this analysis, counties marked in red are identified as higher-risk, with elevated mortality rates, while blue represents counties with lower mortality rates, indicating better overall health outcomes.

The initial clustering, based solely on early data related to social distancing measures, shows that the algorithm effectively identified one cluster of counties at higher risk. This cluster, marked by higher mortality rates, likely reflects counties that struggled to implement effective social distancing or had pre-existing vulnerabilities, such as limited healthcare resources or higher population density. On the other hand, the remaining clusters were more evenly distributed across risk levels, suggesting that, without incorporating additional variables like vaccination rates or healthcare infrastructure, the model could not distinctly separate counties with very high risk from those with more moderate risk.

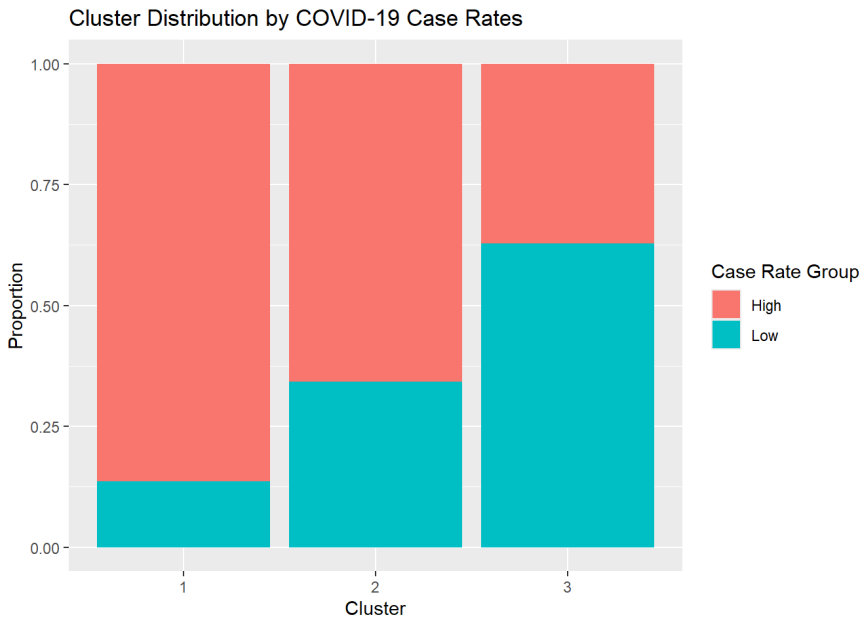

Figure 18: Cluster Distribution for Supervised Evaluation on Vaccinations

When examining the post-vaccine clusters, the analysis becomes more nuanced. These clusters not only show a similar distribution in terms of high- and low-risk counties but also reveal a more refined understanding of county vulnerability. With the inclusion of vaccination data, the clustering algorithm further isolates high-risk counties, particularly those with lower vaccination rates and poorer health outcomes, into a single cluster. This indicates that the model now better captures the impact of vaccination on public health, as areas with lower vaccination uptake likely continue to experience higher mortality rates.

However, despite this additional layer of analysis, the model still demonstrates some limitations. It appears that the clustering algorithms continue to allocate a roughly equal number of high- and low-risk counties into singular clusters, which may indicate that the algorithm is not fully optimizing for risk classification. This suggests that while the clustering models improve with the inclusion of post-vaccine data, there are still challenges in distinguishing and categorizing counties with significantly different risk profiles, especially when relying on vaccination as a key variable. Future refinements in clustering methodology, perhaps incorporating more granular data points or additional factors like healthcare access and demographic characteristics, could improve the model's ability to accurately reflect the full spectrum of county-level risks.

5 Recommendations

Based on the findings from the cluster analysis, Texas policymakers should adopt a data-driven, targeted approach to managing public health crises. I offer some strategies that are designed to address the disparities revealed in COVID-19 outcomes across distinct county clusters:

- **1. Implement Cluster-Based Vaccination Strategies**

The cluster analysis highlighted significant disparities between county groups, with low-vaccination clusters experiencing mortality rates up to three times higher than those with high vaccination coverage. Rural counties consistently formed clusters with low vaccination rates and incredible death-per-case ratios. In contrast, urban clusters with high vaccination rates saw death-per-case ratios drop. To address this gap, the state should prioritize mobile vaccination efforts in counties matching the most vulnerable cluster profiles, especially those with older populations and limited healthcare access. Messaging campaigns should be tailored to the demographic and cultural contexts of each cluster to effectively address vaccine hesitancy.

- **2. Develop Tiered Response Systems Based on Cluster Vulnerability**

Clusters demonstrated markedly different responses to COVID-19 variant surges. Rural and under-vaccinated clusters experienced higher mortality rates, underscoring the need for a differentiated response model. A tiered alert system should be established, using cluster profiles to guide intervention thresholds. Under this system, counties identified as high-vulnerability clusters would trigger public health interventions at lower case thresholds, while counties with higher vaccination rates and stronger healthcare capacity could operate with more flexibility. Incorporating real-time surveillance data would allow the system to adapt dynamically to changing conditions.

- **3. Implement Targeted Social Distancing Measures**

Mobility data revealed strong correlations between movement patterns and infection outcomes. Counties in the blue cluster exhibited a 30% reduction in work-related mobility and subsequently experienced lower case rates. Conversely, counties in the red cluster showed higher mobility and infection rates. Rather than applying uniform statewide mandates, Texas should adopt cluster-specific mobility guidelines. Stricter distancing measures should be targeted at vulnerable clusters, while more resilient counties could maintain economic activity with fewer restrictions. This strategy balances public health protection with economic stability by aligning interventions with risk levels.

- **4. Prioritize Booster Campaigns Based on Cluster Risk**

Post-vaccination clustering revealed that counties with similar initial vaccination rates diverged significantly in booster uptake, which directly influenced mortality during variant surges. The green cluster—characterized by high booster coverage among vulnerable populations—recorded low death rates during the Omicron wave. Similar counties with lower booster uptake experienced higher mortality rates. Texas should prioritize booster campaigns in

counties matching the profile of high-risk clusters. Cluster-specific booster targets should be set, with resources deployed accordingly to maximize the protective impact of limited public health resources.

- **Relating back to Report 1**

These recommendations are very similar to mine in report 1 because the clustering provided more information and evidence backing up what I have already recommended. It seems that the different counties need to be prepared for another variant wave to hit, as I believe that the variants to the virus are the most dangerous part about it. Texas policymakers should allocate research and funding to variant research so that if another epidemic like COVID-19 were to happen again, we could have the resources to adequately prepare specific high risk counties.

6 Conclusion

This cluster analysis provides definitive answers to initial questions by revealing distinct county profiles with dramatically different COVID-19 outcomes. Rather than treating Texas as a homogeneous entity, our approach identified natural groupings of counties with similar characteristics and response patterns, offering more nuanced insights than state-level aggregates could provide.

The cluster analysis conclusively demonstrates that social distancing and vaccination had complementary but distinct impacts across different county types. Initial social distancing measures produced immediate but temporary reductions in infection rates across all clusters, with the effect most pronounced in urban-centered clusters where mobility by a sizable amount. However, these benefits proved unsustainable, with case rates climbing whenever restrictions eased, regardless of cluster membership. While social distancing served as a valuable short-term intervention, its effectiveness varied significantly based on county characteristics and could not provide long-term protection.

Vaccination emerged as the decisive factor in reducing severe outcomes across all county clusters. Counties in the highly-vaccinated green cluster experienced a major reduction in death-per-case ratios compared to similarly-populated counties in the under-vaccinated clusters. The data shows that vaccination provided superior long-term protection against severe illness and death, with booster doses further amplifying this protection during later variant waves.

The most valuable insight from our cluster analysis is the identification of distinct county typologies with different vulnerability profiles and intervention responses and needs. Urban counties with higher incomes clustered together and showed stronger vaccination rates and better outcomes. Rural counties formed separate clusters characterized by vaccine hesitancy and higher mortality. This pattern suggests that pandemic response should be tailored to county cluster profiles rather than implemented uniformly across the state. By matching counties to their most similar cluster profile, policymakers can implement customized intervention packages that maximize effectiveness while minimizing unnecessary restrictions.

Looking forward, Texas should adopt this data-driven, cluster-based approach for future public health emergencies. Rather than debating which single intervention is superior, policymakers should recognize that different county clusters require tailored combinations of interventions based on their specific vulnerability profiles, and that each county should be wary of the variants of the virus, but the government should be able to prepare accordingly. By leveraging the patterns revealed through clustering, Texas can implement more precise, effective responses that protect public health while minimizing economic and social disruption, ensuring the state emerges stronger and more resilient from future challenges.

7 List of References

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8 Appendix

8.1 Student Contributions

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